US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12397146
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Hart; Matthew et al.

Flexible tip for transvalvular circulatory support device

Abstract

Various aspects of the present disclosure are directed towards apparatuses, systems, and methods that may include a blood pump. The blood pump may include a flexible distal tip having a distal end segment, an intermediate segment, and a proximal segment, each segment having various shapes and/or hardness. The blood pump may include a cannula and a can element coupling the flexible distal tip to the cannula.

Inventors: **Hart; Matthew (Maple Grove, MN), Jancaric; Thomas P. (Maple Grove, MN), Anand; Umang (Plymouth, MN), Frost; Matthew N. (New Brighton, MN)**

Applicant: **Boston Scientific Scimed Inc.** (Maple Grove, MN)

Family ID: **1000008779793**

Assignee: **Boston Scientific Scimed Inc. (Maple Grove, MN)**

Appl. No.: **17/720663**

Filed: **April 14, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20220331577 A1	Oct. 20, 2022

Related U.S. Application Data

us-provisional-application US 63174810 20210414

Publication Classification

Int. Cl.: **A61M60/13** (20210101); **A61M25/00** (20060101); **A61M60/174** (20210101)

U.S. Cl.:

CPC **A61M60/13** (20210101); **A61M60/174** (20210101); A61M25/0074 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5527159	12/1995	Bozeman, Jr. et al.	N/A	N/A
5692882	12/1996	Bozeman, Jr. et al.	N/A	N/A
5947892	12/1998	Benkowski et al.	N/A	N/A
6135729	12/1999	Aber	N/A	N/A
6139487	12/1999	Siess	N/A	N/A
6176822	12/2000	Nix et al.	N/A	N/A
6974436	12/2004	Aboul-Hosn et al.	N/A	N/A
7841976	12/2009	McBride et al.	N/A	N/A
7927068	12/2010	McBride et al.	N/A	N/A
7972122	12/2010	LaRose et al.	N/A	N/A
8376707	12/2012	McBride et al.	N/A	N/A
8591393	12/2012	Walters et al.	N/A	N/A
8597170	12/2012	Walters et al.	N/A	N/A
8617239	12/2012	Reitan	N/A	N/A
8684904	12/2013	Campbell et al.	N/A	N/A
8721517	12/2013	Zeng et al.	N/A	N/A
8992163	12/2014	McBride et al.	N/A	N/A
9072825	12/2014	Pfeffer et al.	N/A	N/A
9138518	12/2014	Campbell et al.	N/A	N/A
9162017	12/2014	Evans et al.	N/A	N/A
9308302	12/2015	Zeng	N/A	N/A
9308304	12/2015	Peters et al.	N/A	N/A
9327067	12/2015	Zeng et al.	N/A	N/A
9364592	12/2015	McBride et al.	N/A	N/A
9364593	12/2015	McBride et al.	N/A	N/A
9381288	12/2015	Schenck et al.	N/A	N/A
9421311	12/2015	Tanner et al.	N/A	N/A
9446179	12/2015	Keenan et al.	N/A	N/A
9669142	12/2016	Spanier et al.	N/A	N/A
9675740	12/2016	Zeng et al.	N/A	N/A
9717833	12/2016	McBride et al.	N/A	N/A
9770543	12/2016	Tanner et al.	N/A	N/A
9814814	12/2016	Corbett et al.	N/A	N/A
9872947	12/2017	Keenan et al.	N/A	N/A
9872948	12/2017	Siess	N/A	N/A
9895476	12/2017	LaRose et al.	N/A	N/A
9907890	12/2017	Muller	N/A	N/A

9956332	12/2017	LaRose et al.	N/A	N/A
9962475	12/2017	Campbell et al.	N/A	N/A
9964115	12/2017	Scheckel	N/A	N/A
10029037	12/2017	Muller et al.	N/A	N/A
10039872	12/2017	Zeng et al.	N/A	N/A
10071192	12/2017	Zeng	N/A	N/A
10086121	12/2017	Fitzgerald et al.	N/A	N/A
10105475	12/2017	Muller	N/A	N/A
10117980	12/2017	Keenan et al.	N/A	N/A
10149932	12/2017	McBride et al.	N/A	N/A
10215187	12/2018	McBride et al.	N/A	N/A
10232099	12/2018	Peters et al.	N/A	N/A
10251985	12/2018	Larose et al.	N/A	N/A
10251986	12/2018	Larose et al.	N/A	N/A
10279095	12/2018	Aboul-Hosn et al.	N/A	N/A
10478539	12/2018	Pfeffer et al.	N/A	N/A
10478540	12/2018	Scheckel et al.	N/A	N/A
10525178	12/2019	Zeng	N/A	N/A
10569050	12/2019	Heesch	N/A	N/A
10576192	12/2019	Muller et al.	N/A	N/A
10576193	12/2019	Tanner et al.	N/A	N/A
10709829	12/2019	Muller	N/A	N/A
10709830	12/2019	Tanner et al.	N/A	N/A
10765789	12/2019	Zeng et al.	N/A	N/A
10780208	12/2019	Siess et al.	N/A	N/A
10786610	12/2019	Zeng	N/A	N/A
10799624	12/2019	Pfeffer et al.	N/A	N/A
10842921	12/2019	Siess et al.	N/A	N/A
10864308	12/2019	Muller et al.	N/A	N/A
10864309	12/2019	McBride et al.	N/A	N/A
10874783	12/2019	Pfeffer et al.	N/A	N/A
10894115	12/2020	Pfeffer et al.	N/A	N/A
10918774	12/2020	Stanfield et al.	N/A	N/A
10960116	12/2020	Campbell et al.	N/A	N/A
10980927	12/2020	Pfeffer et al.	N/A	N/A
11058865	12/2020	Fitzgerald et al.	N/A	N/A
11123539	12/2020	Pfeffer et al.	N/A	N/A
11129978	12/2020	Pfeffer et al.	N/A	N/A
11167124	12/2020	Pfeffer et al.	N/A	N/A
11229786	12/2021	Zeng et al.	N/A	N/A
11235138	12/2021	Gross-Hardt et al.	N/A	N/A
11253693	12/2021	Pfeffer et al.	N/A	N/A
11260213	12/2021	Zeng et al.	N/A	N/A
11273301	12/2021	Pfeffer et al.	N/A	N/A
11285309	12/2021	Tuval et al.	N/A	N/A
11305105	12/2021	Corbett et al.	N/A	N/A
11311712	12/2021	Zeng et al.	N/A	N/A
11338124	12/2021	Pfeffer et al.	N/A	N/A
11357967	12/2021	Zeng et al.	N/A	N/A
11400276	12/2021	Chopra et al.	N/A	N/A

11497896	12/2021	Tanner et al.	N/A	N/A
11517736	12/2021	Earles et al.	N/A	N/A
11583659	12/2022	Pfeffer et al.	N/A	N/A
D980971	12/2022	Zarins et al.	N/A	N/A
D980972	12/2022	Zarins et al.	N/A	N/A
11628294	12/2022	Chopra et al.	N/A	N/A
11708833	12/2022	McBride et al.	N/A	N/A
11754075	12/2022	Schuelke et al.	N/A	N/A
11786700	12/2022	Pfeffer et al.	N/A	N/A
2003/0187322	12/2002	Siess	N/A	N/A
2006/0161095	12/2005	Aboul-Hosn et al.	N/A	N/A
2008/0103591	12/2007	Siess	N/A	N/A
2008/0114339	12/2007	McBride et al.	N/A	N/A
2009/0060743	12/2008	McBride et al.	N/A	N/A
2009/0171137	12/2008	Farnan et al.	N/A	N/A
2013/0303831	12/2012	Evans	N/A	N/A
2017/0296725	12/2016	Peters et al.	N/A	N/A
2018/0110908	12/2017	Larose	N/A	N/A
2018/0303990	12/2017	Siess et al.	N/A	N/A
2018/0311423	12/2017	Zeng et al.	N/A	N/A
2019/0344001	12/2018	Salahieh et al.	N/A	N/A
2019/0351180	12/2018	Ryu	N/A	A61B 34/20
2020/0330664	12/2019	Ship	N/A	A61M 60/135
2021/0015982	12/2020	Kerkhoffs et al.	N/A	N/A
2021/0038785	12/2020	Siess et al.	N/A	N/A
2021/0038791	12/2020	Tuval et al.	N/A	N/A
2021/0106810	12/2020	Pfeffer et al.	N/A	N/A
2022/0134082	12/2021	Pfeffer et al.	N/A	N/A
2022/0184372	12/2021	Tuval et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
0847767	12/2004	EP	N/A
2301598	12/2016	EP	N/A
3352808	12/2022	EP	N/A

OTHER PUBLICATIONS

US 9,607,007, 06/2015, Tanner et al. (withdrawn) cited by applicant
International Search Report and Written Opinion for International Application No. PCT/US2022/024782 dated Jul. 28, 2022 (14 pages). cited by applicant

---

Primary Examiner: Layno; Carl H

Assistant Examiner: Lee; Bryan McAllister

Attorney, Agent or Firm: Seager, Tufte & Wickhem LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application claims priority to Provisional Application No. 63/174,810, filed Apr. 14, 2021, which is herein incorporated by reference in its entirety.

TECHNICAL FIELD

(1) The present disclosure relates to percutaneous circulatory support devices. More specifically, the disclosure relates to flexible tips for use in percutaneous circulatory support devices.

BACKGROUND

(2) Percutaneous circulatory support devices such as blood pumps can provide transient support for hours or months of use in patients whose heart function or cardiac output is compromised. Such devices are delivered percutaneously from the femoral artery, retrograde through the descending aorta, over the aortic arch, through the ascending aorta across the aortic valve, and into the left ventricle. Several issues may complicate the delivery and operation of blood pumps within the heart, including chordae tendineae entanglement, difficulty with guidewire movement, interference with lateral left ventricle contraction, generation of premature ventricular contractions, trauma to cardiac tissue, and oscillation of the blood pump resulting in decreased performance of the blood pump.

SUMMARY

(3) In Example 1, a blood pump comprises a flexible distal tip comprising a distal end segment, an intermediate segment, and a proximal segment. The intermediate segment has a curved shape, and the distal end segment has a curved shape that overlaps with the intermediate segment but not the proximal segment.

(4) In Example 2, the blood pump of Example 1, wherein the proximal segment has a first hardness, the distal end segment has a second hardness, and the first hardness is greater than the second hardness.

(5) In Example 3, the blood pump of any of Examples 1-2, wherein the flexible distal tip comprises a polymer.

(6) In Example 4, the blood pump of any of Examples 1-3, wherein the proximal segment has a first wall thickness and the distal end segment has a second wall thickness.

(7) In Example 5, the blood pump of any of Examples 1-4, wherein the proximal segment is comprised of a first material and the distal end segment is comprised of a second material.

(8) In Example 6, the blood pump of any of Examples 1-5, further comprising a cannula and a can element. The can element couples the flexible distal tip to the cannula.

(9) In Example 7, the blood pump of Example 6, wherein the can element comprises a distal component and a proximal component.

(10) In Example 8, the blood pump of any of Examples 6-7, wherein the cannula comprises one or more tines, and wherein the one or more tines are coupled to the proximal component of the can element with an adhesive.

(11) In Example 9, the blood pump of any of Examples 6-8, wherein the distal tip is coupled to the can element with an adhesive.

(12) In Example 10, a blood pump comprises a cannula and a flexible distal tip forming a loop. The loop has a proximal end coupled to the cannula, a distal end, and a lumen extending between the proximal end and the distal end.

(13) In Example 11, the blood pump of Example 10, wherein the lumen allows for passage of a guidewire from the proximal end of the loop to the distal end of the loop.

(14) In Example 12, the blood pump of any of Examples 10-11, wherein the lumen extends from the proximal end of the loop to the distal end of the loop via a guidewire support lumen connecting the proximal end and the distal end of the loop.

(15) In Example 13, the blood pump of any of Examples 10-12, wherein the flexible distal tip comprises a polymer.

(16) In Example 14, the blood pump of any of Examples 10-13, wherein the flexible distal tip has a length of approximately 20 mm and a width of approximately 10 mm.
(17) In Example 15, the blood pump of any of Examples 10-14, wherein the flexible distal tip further comprises a secondary loop. The loop is joined to the secondary loop such that the loop is offset from the secondary loop by 90 degrees.
(18) In Example 16, a blood pump comprises a flexible distal tip comprising a distal end segment, an intermediate segment, and a proximal segment. The proximal segment has a relatively straight shape, the intermediate segment has a curved shape, and the distal end segment has a curved shape that overlaps with the intermediate segment but not the proximal segment.
(19) In Example 17, the blood pump of Example 16, wherein the proximal segment has a first hardness, the distal end segment has a second hardness, and the first hardness is greater than the second hardness.
(20) In Example 18, the blood pump of Example 16, wherein the flexible distal tip comprises a polymer.
(21) In Example 19, the blood pump of Example 16, wherein the proximal segment has a first wall thickness and the distal end segment has a second wall thickness.
(22) In Example 20, the blood pump of Example 16, wherein the proximal segment is comprised of a first material and the distal end segment is comprised of a second material.
(23) In Example 21, the blood pump of Example 16, further comprising a cannula and a can element. The can element couples the flexible distal tip to the cannula.
(24) In Example 22, the blood pump of Example 21, wherein the can element comprises a distal component and a proximal component.
(25) In Example 23, the blood pump of Example 21, wherein the cannula comprises one or more tines, and wherein the one or more tines are coupled to the proximal component of the can element with an adhesive.
(26) In Example 24, the blood pump of Example 21, wherein the distal tip is coupled to the can element with an adhesive.
(27) In Example 25, a blood pump comprises a cannula and a flexible distal tip forming a loop. The loop has a proximal end coupled to the cannula, a distal end, and a lumen extending between the proximal end and the distal end.
(28) In Example 26, the blood pump of Example 25, wherein the lumen allows for passage of a guidewire from the proximal end of the loop to the distal end of the loop.
(29) In Example 27, the blood pump of Example 26, wherein the lumen extends from the proximal end of the loop to the distal end of the loop via a curved segment of the loop.
(30) In Example 28, the blood pump of Example 26, wherein the lumen extends from the proximal end of the loop to the distal end of the loop via a guidewire support lumen connecting the proximal end and the distal end of the loop.
(31) In Example 29, the blood pump of Example 25, wherein the flexible distal tip comprises a polymer.
(32) In Example 30, the blood pump of Example 25, wherein the flexible distal tip has a length of approximately 20 mm and a width of approximately 10 mm.
(33) In Example 31, the blood pump of Example 25, wherein the flexible distal tip further comprises a secondary loop. The loop is joined to the secondary loop such that the loop is offset from the secondary loop by 90 degrees.
(34) In Example 32, the blood pump of Example 31, wherein the secondary loop comprises a polymer.
(35) In Example 33, a method of assembling a blood pump comprises coupling one or more tines to a cannula, coupling the one or more tines to a proximal component of a can element, coupling a distal component of a can element to the proximal component of the can element, and coupling a flexible distal tip to the distal component of the can element

(36) In Example 34, the blood pump of Example 33, further comprising coupling the distal tip to the can element using an adhesive.
(37) In Example 35, the blood pump of Example 33, further comprising coupling the distal component to the proximal component using an adhesive.
(38) While multiple embodiments are disclosed, still other embodiments of the present invention will become apparent to those skilled in the art from the following detailed description, which shows and describes illustrative embodiments of the invention. Accordingly, the drawings and detailed description are to be regarded as illustrative in nature and not restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** depicts a portion of an illustrative percutaneous mechanical circulatory support device (also referred to herein, interchangeably, as a “blood pump”), and its relative position in a heart, in accordance with embodiments of the subject matter disclosed herein.
(2) FIG. **2**A depicts a side view of a portion of an embodiment of an illustrative percutaneous mechanical circulatory support device, in accordance with embodiments of the subject matter disclosed herein.
(3) FIG. **2**B depicts an enlarged view of a portion of the embodiment of an illustrative percutaneous mechanical circulatory support device shown in FIG. **2**A, in accordance with embodiments of the subject matter disclosed herein.
(4) FIG. **3**A depicts a side view of a portion of an embodiment of an illustrative percutaneous mechanical circulatory support device, in accordance with embodiments of the subject matter disclosed herein.
(5) FIG. **3**B depicts a side view of a portion of an embodiment of an illustrative percutaneous mechanical circulatory support device, in accordance with embodiments of the subject matter disclosed herein.
(6) FIG. **3**C depicts a side view of a portion of an embodiment of an illustrative percutaneous mechanical circulatory support device, in accordance with embodiments of the subject matter disclosed herein.
(7) FIG. **4** depicts a side view of a portion of an alternative embodiment of an illustrative percutaneous mechanical circulatory support device, in accordance with embodiments of the subject matter disclosed herein.
(8) While the invention is amenable to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and are described in detail below. The intention, however, is not to limit the invention to the particular embodiments described. On the contrary, the invention is intended to cover all modifications, equivalents, and alternatives falling within the scope of the invention as defined by the appended claims.
DETAILED DESCRIPTION
(9) FIG. **1** depicts a portion of an illustrative percutaneous mechanical circulatory support device (also referred to herein, interchangeably, as a “blood pump”), and its relative position in a human heart, in accordance with embodiments of the subject matter disclosed herein. Blood pump **100** may be delivered percutaneously using a guidewire (not shown), passing from the femoral artery (not shown), retrograde through the descending aorta (not shown), over the aortic arch (not shown), through the ascending aorta **106**, past the aortic valve **110**, and into the left ventricle **112**. Blood pump **100** includes a cannula **114** and a flexible distal tip **108**. The cannula **114** may also have a flexible construction to facilitate delivery of the blood pump **100**. The cannula **114** includes one or more blood inlets **116** located on a distal section **118** of the cannula **114**, and one or more blood outlets **120** which are located within the pump housing **122**.

(10) During delivery, the blood pump **100**, and in particular the distal tip **108**, may contact structures within the heart, such as the aortic valve **110**, one or more chordae tendineae **124** emanating from the posterior papillary muscle **102** or anterior papillary muscle **104**. During operation, the blood pump **100** is positioned within the heart such that the one or more blood inlets **116** is positioned in the left ventricle **112** and the one or more blood outlets **120** is positioned in the aorta **106**, as show in FIG. **1**. The blood pump **100** may be positioned such that the distal tip **108** contacts the left ventricular wall **126**, for example, in the location of the apex **128** of the left ventricle **112**.

(11) FIG. **2**A depicts a side view of a portion of an embodiment of an illustrative percutaneous mechanical circulatory support device, in accordance with embodiments of the subject matter disclosed herein. As shown, a blood pump **200** includes a cannula **210**, a can element **212**, one or more tines **214** (described in more details with respect to FIG. **2**B), and a flexible distal tip **208**. The distal tip **208** includes a distal end segment **202**, an intermediate segment **204**, and a proximal segment **206**. The distal tip **208** also may include a lumen (not shown) to facilitate passage of a guidewire (not shown). The proximal segment **206** has a relatively straight shape, the intermediate segment **204** has a curved shape, and the distal end segment **202** has a curved shape that overlaps with the intermediate segment **204** but not the proximal segment **206**. In some embodiments, and as shown in FIG. **2**A, a bend segment **207** may extend and form a gradually curved transition between proximal segment **206** and intermediate segment **204**. A guidewire (not shown) may extend through the cannula **210**, can element **212**, and distal tip **208** via the lumen of the distal tip **208**. In some embodiments, the guidewire diameter may be approximately 0.018 inches. In order to accommodate the guidewire, the inner diameter of the lumen of the distal tip **208** may be from 0.030 to 0.050 inches, and the outer diameter of the lumen of the distal tip **208** may be from 0.050 to 0.100 inches. In certain embodiments, the inner diameter of the lumen of the distal tip **208** may be approximately 0.040 inches, and the outer diameter of the lumen of the distal tip **208** may be approximately 0.080 inches.

(12) The distal tip **208** may be made of one or more materials, such as one or more polymers (i.e. Pebax, Pellethane), rubbers, plastics, or other materials known to a person skilled in the art that allow the distal tip **208** to be flexible, relatively soft, and/or collapsible. In some embodiments, a preferred material for the distal tip **208** is Pebax. In certain embodiments, the distal tip **208** may be made of a flexible metal material (e.g., Nitinol, Elgiloy, or another metal material with desirable properties). Distal tip **208** may also include features, such as slots or other structures to improve flexibility, collapsibility, performance, durability, or other desirable characteristics. In some embodiments, the distal tip **208** may be slotted and made of Nitinol that may be heat set to various shapes. In certain embodiments, and in particular when the distal tip **208** has a tighter radius (i.e. smaller overall diameter) or when the distal tip **208** includes long, curved lengths, the inner diameter of the distal tip **208** may be lined with a lubricious material or liner to facilitate passage of the guidewire through the lumen of the distal tip **208**.

(13) Introduction of a guidewire (not shown) into the distal tip **208** may expand or straighten the curved shapes of the intermediate segment **204** and distal end segment **202** of the distal tip **208**. In certain instances, when the guidewire is retracted, the curved shape of the intermediate segment **204** and distal end segment **202** may retract to their original shape. In particular, choice of the shape and material of the distal tip **208** may allow for the tip **208** to unfurl during extraction of the blood pump **200** and disengage from the one or more chordae or other structures without tearing or damaging these structures. In other instances, depending on the material used for the flexible distal tip **208**, when the guidewire is retracted, the curved shape of the intermediate segment **204** and distal end segment **202** may not retract to their original shape, and may remain expanded or straightened.

(14) The proximal segment **206** of the distal tip **208** is constructed to have moderate stiffness, so that it may act as a dampener absorbing forces acting on the distal tip **208** and blood pump **200**.

Moderate stiffness for the proximal segment **206** is also beneficial so that the distal tip **208** may provide axial strength, which is advantageous for positioning and supporting the cannula **210** in the left ventricle. The appropriate stiffness of proximal segment **206** may be achieved by constructing the proximal segment **206** of one or more materials of appropriate hardness, by the inclusion of structures, such as reinforcement structures or slots, within the proximal segment **206** to achieve the appropriate stiffness, by combining materials and structures to achieve the appropriate stiffness, or by using other techniques known to one of ordinary skill in the art. In some advantageous embodiments, the bend segment **207** may also facilitate dampening and absorbing axial forces acting on the distal tip **208**, and be constructed of similar materials as the proximal segment **206**.

(15) The intermediate segment **204** of the distal tip **208** is constructed to have a stiffness less than the stiffness of the proximal segment **206**. Such a stiffness for the intermediate segment **204** is beneficial so that the intermediate segment **204** does not cause trauma or damage when contacting tissue, yet provides adequate structural strength for positioning and supporting of the cannula **210** in the left ventricle while also being capable of absorbing forces acting on the distal tip **208**. The appropriate stiffness of intermediate segment **204** may be achieved by constructing the intermediate segment **204** of one or more materials of appropriate hardness, by the inclusion of structures, such as reinforcement structures or slots, within the intermediate segment **204** to achieve the appropriate stiffness, by combining materials and structures to achieve the appropriate stiffness, or by using other techniques known to one of ordinary skill in the art. In general, the intermediate segment **204** may be constructed of materials that have a hardness less than the hardness of the materials forming the proximal segment **206**, as measured, for example, by a durometer.

(16) The distal end segment **202** of the distal tip **208** is constructed to have a stiffness less than the stiffness of the intermediate segment **204**. Such a stiffness for the distal end segment **202** is beneficial so that distal end segment **202** does not cause trauma or damage when contacting tissue, yet provides adequate structural strength for positioning and supporting of the cannula **210** in the left ventricle while also being capable of absorbing forces acting on the distal tip **208**. The appropriate stiffness of the distal end segment **202** may be achieved by constructing the distal end segment **202** of one or more materials of appropriate hardness, by the inclusion of structures, such as reinforcement structures or slots, within the distal end segment **202** to achieve the appropriate stiffness, by combining materials and structures to achieve the appropriate stiffness, or by using other techniques known to one of ordinary skill in the art. In general, the distal end segment **202** may be constructed of materials that have a hardness less than the hardness of the materials forming intermediate segment **204**, as measured, for example, by a durometer. In certain embodiments, based on the materials used for the reinforcement structures, the inclusion of the structures may aid in the visualization for the distal tip **208** under fluoroscopy.

(17) As described above, the distal tip **208** has a proximal segment **206** and a distal end segment **202**, and the hardness of the proximal segment **206** may be greater than the hardness of the distal end segment **202**. In certain embodiments, the hardness transition may occur in discrete steps along the length of the distal tip **208**. In some embodiments, the hardness transition may be gradual or continuous along the length of the distal tip **208**. In other embodiments, the hardness transition may be a combination of discrete steps and continuous segments. The decrease in hardness from the proximal segment **206** to the distal end segment **202** of the distal tip **208** may be achieved through a variety of ways, such as decreasing wall thickness of the distal tip **208** from the proximal segment **206** to the distal end segment **202**, decreasing hardness of material along the length of distal tip **208** without using discrete segments (e.g., intermittent layer co-extrusion), changing the pattern or construction of the walls of the distal tip **208** along the length of distal tip **208**, changing the pattern or construction of internal support structures within the walls of the distal tip **208** along the length of the distal tip **208**, heat treating the distal tip **208** or portions thereof to change the material properties of the distal tip **208** or portions thereof, or by any other method known by a person of ordinary skill in the art.

(18) There are numerous advantages to the distal tip **208** described in relation to FIG. **2**A. Constructing the distal tip **208** to be flexible, relatively soft, and/or collapsible helps prevent the distal tip **208** from causing trauma or damage to vascular or cardiac tissues and structures during delivery or operation of the blood pump **200**, and may assist in delivery of the blood pump through the vascular and cardiac structures. For example, the curved shapes and material properties of the intermediate segment **204** and the distal end segment **202** protect against potential entanglement of the distal tip **208** in cardiac structures, such as the one or more set of chordae tendineae **124** emanating from the posterior papillary muscle or the anterior papillary muscle (shown in FIG. **1**), during delivery of the blood pump **200**. Similarly, the overlapping arrangement of the intermediate segment **204** and the distal end segment **202** avoids potential entanglement of the distal tip **208** with cardiac structures when introducing the blood pump **200** into the heart. A flexible and/or relatively soft distal tip **208** also helps prevent the guidewire from damaging or causing trauma to vascular or cardiac structures. In addition, a flexible and/or collapsible distal tip **208** facilitates navigation of the blood pump **200** through vascular and cardiac structures. For example, the curved shapes and material properties of the intermediate segment **204** and the distal end segment **202** allow the distal tip **208** to unfurl during extraction of the blood pump **200**, permitting the distal tip **208** to pass through or disengage from the chordae without tearing or damaging these structures. Also, a flexible, relatively soft, and/or collapsible distal tip **208** also helps maintain the blood pump **200** in proper position within the heart by absorbing or dampening forces, such as blood flow or ventricular wall contact, acting on the blood pump **200** when the distal tip **208** is positioned against a wall of the left ventricle. If the cannula **210** is constructed to be flexible, there may be movement or oscillation of the blood pump **200** within the heart due to blood flow and contraction of the heart. The flexible, relatively soft, and/or collapsible distal tip **208** also accounts for and minimizes resistance to lateral contraction forces acting on the tip **208** during contraction of the left ventricle. The distal tip **208** as shown in FIG. **2**A significantly reduces such movement or oscillation by supporting the distal tip **208** of the blood pump **200** against a wall of the left ventricle, thereby stabilizing the overall blood pump **200**. Such stabilization of the blood pump **200** may increase efficiency, performance, and/or longevity of the blood pump **200**.
(19) FIG. **2**B depicts an enlarged view of a portion of the embodiment of an illustrative percutaneous mechanical circulatory support device shown in FIG. **2**A. As shown, the blood pump **200** includes a can element **212**, a cannula **210**, and a flexible distal tip **208** (partially shown). The can element **212** includes a distal component **216** and a proximal component **218**, and may be used to couple the distal tip **208** to the cannula **210**. The cannula **210** further includes one or more tines **214**. The one or more tines **214** may be coupled to the proximal component **218** of the can element **212** using an adhesive or mechanical coupling, preferably prior to the distal component **216** of the can element **212** being attached to the proximal component **218** of the can element **212**. Such a construction provides, for example, a direct line of sight available when curing an adhesive. The distal component **216** of the can element **212** may then be slid over the proximal component **218** of the can element **212** and attached to the proximal component **218** of the can element **212**. The distal component **216** may be mechanically coupled to the proximal component **218** of the can element **212**, or coupled with an adhesive. The distal tip **208** may be coupled to the proximal component **218** of the can element **212** using an adhesive or mechanical coupling, either before or after the distal component **216** of the can element **212** is attached to the proximal component **218** of the can element **212**. The distal tip **208** may also be coupled to the distal component **216**, or the proximal component **218** and distal component **216**, of the can element **212**.
(20) In some embodiments, the can element **212** may be composed of stainless steel and the distal tip **208** may be composed of Nitinol. In such embodiments, the distal tip **208** may be joined to the can element **212** using an adhesive, as the use of an adhesive is a preferred method for joining dissimilar materials, and laser or other types of welding are not preferred for joining such materials. One preferred example of suitable adhesive for joining the proximal component **218** and distal

component **216** is ultraviolet adhesive. The use of ultraviolet adhesives results in an average bond strength of 6 pounds per foot. Laser or other types of welding may be used in other embodiments, and may be preferable, for joining similar metals or other materials. For example, in other embodiments, the distal tip **208** and can element **212** may be made of similar metals, and may be joined together using laser welding, or other techniques known by a person of ordinary skill in the art. Alternative methods and materials known to one of ordinary skill in the art for securing the tines **214** to the cannula **210**, the distal tip **208** to the proximal component **218** and/or distal component **216** of the can element **212**, or the proximal component **218** and distal component **216** of the can element **212**, may be used. Examples of alternative methods include the use of welding, soldering, brazing, crimping, threaded connection, and other types of mechanical couplings. In certain embodiments, the use of mechanical couplings may be preferred to achieve consistency in strength and durability.

(21) The above-described can element **212** provides several advantages. For example, inclusion of the can element **212** provides mass at the distal section of the cannula **210**, which contributes to the overall cannula and blood pump stability. Constructing the can element **212** of a highly polished stainless-steel material may also minimize thrombus formation on the blood pump. The overall design of the can element **212** and distal tip **208** also provides a seamless transition from the proximal component **218** of the can element **212** into the distal tip **208**, helping to prevent a guidewire (not shown here) from catching on the proximal end of the distal tip **208** or components of the can element **212**. The above-described can element **212** is compatible with alternative embodiments of distal tips discussed below with respect to FIGS. **3**A-**30** and FIG. **4**.

(22) FIG. **3**A depicts a side view of a portion of an alternative embodiment of an illustrative percutaneous mechanical circulatory support device, in accordance with embodiments of the subject matter disclosed herein. As shown, a partial view of a blood pump **300** includes a can element **302**, a center lumen **304**, and a flexible distal tip **308**. The distal tip **308** forms a closed loop **309**, which includes a proximal end **310** and a distal end **312**. The closed loop shape allows for the distal tip **308** to perform its beneficial functions (e.g., stabilization of the blood pump, facilitating guidewire movement, reducing the potential for chordae entanglement, etc.). In embodiments, the inner diameter of the center lumen **304** may be from 0.030 to 0.050 inches, and the outer diameter of the center lumen **304** may be from 0.050 to 0.100 inches. In some embodiments, the inner diameter of the center lumen **304** may be approximately 0.04 inches, and the outer diameter of the center lumen **304** may be approximately 0.08 inches. In some embodiments, the length of the closed loop **309** may be around 10-40 mm, and the width of the closed loop **309** may be around 5-20 mm. In some embodiments, the length of the closed loop **309** may be approximately 20 mm, and the width of the closed loop **309** may be approximately 10 mm.

(23) In the embodiment in FIG. **3**A, the center lumen **304** does not extend through curved portions **316**, **318** of the closed loop **309**; instead, as shown, the guidewire **306** passes through the center lumen **304**, and then across the closed loop **309** between the proximal end **310** and distal end **312** of the distal tip **308**. Alternatively, as shown in FIG. **3**B, the center lumen **304** may extend from the proximal end **310** of the loop **309** to the distal end **312** of the loop **309** via one of the curved portions **316** or **318** of the closed loop **309**, such that guidewire **306** may pass from the proximal end **310**, through one of curved portions **316**, **318**, to distal end **312** of the distal tip **308**. In another alternative, as shown in FIG. **3**C, the center lumen **304** may extend from the proximal end **310** of the loop **309** to the distal end **312** of the loop **309** via a guidewire support lumen **314**, which directly connects the proximal end **310** and the distal end **312** of the loop **309**. In other respects, the distal tip **308** shown in FIGS. **3**A, **3**B, and **3**C may be constructed of the materials and according to the dimensions of the embodiment described above with respect to FIG. **2**A. For example, the inner diameter of the distal tip **308** may be lined with a lubricious material or liner to facilitate passage of a guidewire through the lumen of the distal tip **308**.

(24) FIG. **4** depicts a side view of a portion of an alternative embodiment of an illustrative

percutaneous mechanical circulatory support device, in accordance with embodiments of the subject matter disclosed herein. As shown, a partial view of a blood pump **400** includes a can element **402** and a flexible distal tip **408**. The distal tip **408** includes a first loop **404** and a second loop **406**. The first loop **404** is joined to the second loop **406** such that the first loop **404** is offset from the second loop **406** by 90 degrees, forming a 3-dimensional "whisk" shape. This "whisk" shaped distal tip **408** allows for multiple contact points in the left ventricle, which reduces the contact pressure on the ventricle wall at any one point of contact, thereby minimizing the risk of tissue damage. The distal tip **408** shown in FIG. **4** may be constructed of the materials and according to the dimensions of the embodiment described above with respect to FIG. **2**A, including, for example, the use of a lubricious material or liner to facilitate passage of a guidewire through the lumen of the distal tip **308**.

(25) In some embodiments, a guidewire (not shown) may pass through the proximal portion **410** and distal portion **412** of the distal tip **408** without overlapping with the first loop **404** or the second loop **406**. Alternatively, a guidewire may extend through one of the curved portions of one of the loops **404**, **406** in addition to the proximal portion **410** and distal portion **412** of the distal tip **408**. Alternatively, in some embodiments, the distal tip **408** may include a guidewire support lumen (not shown, but similar to what is shown in FIG. **3**C) extending across the loops **404**, **406**. In embodiments, the inner diameter of the guidewire support lumen may be from 0.030 to 0.050 inches, and the outer diameter of the guidewire support lumen may be from 0.050 to 0.100 inches. In some embodiments, the inner diameter of the guidewire support lumen may be approximately 0.04 inches, and the outer diameter of the guidewire support lumen may be approximately 0.08 inches.

(26) Various modifications and additions can be made to the exemplary embodiments discussed without departing from the scope of the present invention. For example, while the embodiments described above refer to particular features, the scope of this invention also includes embodiments having different, combinations of features and embodiments that do not include all of the described features. Accordingly, the scope of the present invention is intended to embrace all such alternatives, modifications, and variations as fall within the scope of the claims, together with all equivalents thereof.

Claims

1. A blood pump, comprising: a flexible distal tip comprising: a distal end segment having a first end and a second end; an intermediate segment having a first end and a second end; and a proximal segment having a first end and a second end; wherein the distal end of the proximal segment adjoins the proximal end of the intermediate segment at a first location, and the distal end of the intermediate segment adjoins the proximal end of the distal end segment at a second location; and wherein the proximal segment has a relatively straight shape from the proximal end of the proximal segment to the distal end of the proximal segment, the intermediate segment has a curved shape from the proximal end of the intermediate segment to the distal end of the intermediate segment, and the distal end segment has a curved shape that overlaps with the intermediate segment but does not extend proximal of the distal end of the proximal segment.
2. The blood pump of claim 1: wherein the proximal segment has a first hardness, and the distal end segment has a second hardness, and wherein the first hardness is greater than the second hardness.
3. The blood pump of claim 1, wherein the flexible distal tip comprises a polymer.
4. The blood pump of claim 1, wherein the proximal segment has a first wall thickness and the distal end segment has a second wall thickness.
5. The blood pump of claim 1, wherein the proximal segment is comprised of a first material and the distal end segment is comprised of a second material.

6. The blood pump of claim 1, further comprising: a cannula; and a can element; wherein the can element couples the flexible distal tip to the cannula.
7. The blood pump of claim 6, wherein the can element comprises a distal component and a proximal component.
8. The blood pump of claim 6, wherein the cannula comprises one or more tines, and wherein the one or more tines are coupled to the proximal component of the can element with an adhesive.
9. The blood pump of claim 6, wherein the distal tip is coupled to the can element with an adhesive.
10. A method of assembling a blood pump, comprising: coupling one or more tines to a cannula; coupling the one or more tines to a proximal component of a can element; coupling a distal component of a can element to the proximal component of the can element; and coupling a flexible distal tip to the distal component of the can element; wherein the distal component of the can element extends distally past a distal end of the tines.
11. The method of claim 10, further comprising: coupling the distal tip to the can element using an adhesive.
12. The method of claim 10, further comprising: coupling the distal component to the proximal component using an adhesive.
13. A blood pump, comprising: a flexible distal tip comprising: a distal end segment having a first end and a second end; an intermediate segment having a first end and a second end; and a proximal segment having a first end and a second end; wherein the distal end of the proximal segment adjoins the proximal end of the intermediate segment at a first location, and the distal end of the intermediate segment adjoins the proximal end of the distal end segment at a second location; wherein the proximal segment has a relatively straight shape defining a longitudinal axis, the intermediate segment has a curved shape, and the distal end segment has a curved shape that overlaps with the intermediate segment but does not extend proximal of the distal end of the proximal segment, and wherein the intermediate segment curves in a first direction away from the longitudinal axis, and the distal end segment curves in a second direction to intersect with the longitudinal axis.
14. The blood pump of claim 13: wherein the proximal segment has a first hardness, and the distal end segment has a second hardness, and wherein the first hardness is greater than the second hardness.
15. The blood pump of claim 13, wherein the flexible distal tip comprises a polymer.
16. The blood pump of claim 13, wherein the proximal segment has a first wall thickness and the distal end segment has a second wall thickness.
17. The blood pump of claim 13, wherein the proximal segment is comprised of a first material and the distal end segment is comprised of a second material.
18. The blood pump of claim 13, further comprising: a cannula; and a can element; wherein the can element couples the flexible distal tip to the cannula.
19. The blood pump of claim 18, wherein the can element comprises a distal component and a proximal component.
20. The blood pump of claim 18, wherein the cannula comprises one or more tines, and wherein the one or more tines are coupled to the proximal component of the can element with an adhesive.